PAWAN KUMAR

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EDUCATION

University of Bristol : UK — **September 2026 – Present**
MSc - Data Science — *Bristol, United Kingdom*

Indian Institute of Technology, Varanasi — **Nov'2021 – May'2025**
B.Tech - Computer Science and Engineering - ***CPI - 7.99*** — *Varanasi, Uttar Pradesh*

TECHNICAL SKILLS

Programming: C++, C, Python(ML),R.
Technologies: HTML5, CSS3,TensorFlow, MySQL, Bootstrap,Scikit-learn, Keras, Pandas, NumPy.
Tools: VS Code,Linux , Django.

EXPERIENCE

IITRAM — **Sep 2025 – Jan 2026**
Junior Research Fellow (JRF) | *Advisor: Dr. Prasun Chandra Tripathi*

- Developed **NoiseAwareSwinUNet**, a hybrid Swin Transformer and Restormer-based architecture for MRI denoising, integrating MDTA and GDFN blocks to capture global and local features.
- Achieved state-of-the-art restoration quality with a **PSNR of 40.92 dB** and **SSIM of 0.97**, significantly enhancing diagnostic clarity.
- Co-authored a conference paper detailing the novel integration of Swin blocks with Gated-Dconv Feed-Forward Networks for WB-DWI image denoising accepted at NCVPRIPG 2026.

PROJECTS

Online Examination Portal (STA) | HTML, CSS, Django, Python, Database - MySQL **Jul'2024-Sep'2024**

- Designed and enforced a dynamic online examination portal with secure user authentication and CSRF protection.
- Created an admin panel to automate exam scheduling, reducing manual effort by **70%**.
- Engineered responsive user interfaces and integrated MySQL for efficient data storage.

Railway Ticket Booking System | MySQL,HTML,CSS,Javascript,Spring Boot **Jan'2023-Jun'2023**

- Crafted and carried out a comprehensive IRCTC-like ticket booking system using Spring Boot and MySQL.
- Deployed sophisticated database functionalities for managing bookings, cancellations, real-time seat availability, resource-efficient seat allocation, and dynamic handling of waiting lists.
- Merged canceled sub-route seats with adjacent available routes to create longer routes for efficient resource allocation.

Federated Time Series Forecasting | Python, TensorFlow, Machine Learning **Jul'2023-May 2024**

- Devised and implemented a federated learning framework integrating Temporal Convolutional Networks (TCN) with Bidirectional LSTM to accurately forecast energy usage in North and South regions.
- Achieved an R^2 score of **0.999997 and MSE of 0.0012**, demonstrating exceptional predictive accuracy.
- Built data preprocessing pipelines, including normalization and feature extraction, ensuring efficient handling of time-series data across clients.

Achievements

- Co-authored a pre-print on deep learning for medical imaging on arXiv (June 2026) and paper accepted at **NCVPRIPG 2026.**
- Placement Offer from **Morabu Hanshin** Japan based IT firm.
- Solved **500+** problems across **Codeforces** and **GeeksforGeeks**.

CERTIFICATIONS

- Data Structures and Algorithms in C and C++ - Udemy.
- Machine Learning- Udemy.

EXTRACURRICULAR

Student Counselling Service — **Nov'2022–Jan'2023**
Academic Mentor - MA102

- Guided and supported first-year students in navigating academic challenges, enhancing their understanding of course material, and effectively preparing for exams.